

Lecture 26: Characterizations of Trees and Spanning Trees

Last time, we defined a tree as a connected graph with no cycles. Today we prove several equivalent ways to recognize trees and then use spanning trees in two applications.

Let G be a finite graph with

$$n = |V(G)|, \quad m = |E(G)|.$$

Recall:

tree = connected + acyclic.

We will prove that the following three conditions are equivalent:

- (1) G is connected and acyclic,
- (2) G is connected and $m = n - 1$,
- (3) G is acyclic and $m = n - 1$.

The proof will follow the cycle

$$(1) \implies (2) \implies (3) \implies (1).$$

Once this cycle is proved, any one of the three conditions implies the other two. As a result, each condition gives a standalone test for being a tree.

From trees to the edge count. We already proved the first implication last lecture:

Every tree with n vertices has $n - 1$ edges.

The proof used the longest path trick: the endpoints of a longest path in a tree are leaves. Then we removed a leaf, applied induction to the smaller tree, and added the leaf back.

Therefore, if G is connected and acyclic, then G is a tree, so $m = n - 1$. Hence

$$(1) \implies (2).$$

Connected with $n - 1$ edges means acyclic. Now assume that G is connected and $m = n - 1$. We prove that G is acyclic.

Suppose, to the contrary, that G contains a cycle. Deleting one edge from a cycle does not disconnect the graph: the two endpoints of the deleted edge are still connected by the rest of the cycle.

Starting from G , delete edges from cycles one at a time until no cycles remain. Suppose we delete k edges. Since we assumed that there was at least one cycle, we have $k \geq 1$. At every step, the graph stays connected. The final graph is connected and acyclic, so it is a tree on the same n vertices; consequently, the final graph must have $n - 1$ edges. On the other hand, we started with $n - 1$ edges and deleted k of them, so the final number of edges is

$$(n - 1) - k.$$

Equating the two expressions,

$$(n - 1) - k = n - 1,$$

which forces $k = 0$. This contradicts $k \geq 1$. Hence G had no cycle to begin with, so G is acyclic. This proves the implication

$$(2) \implies (3).$$

Acyclic with $n - 1$ edges means connected. A **forest** is an acyclic graph. Equivalently, a forest is a disjoint union of trees.

Let G be a forest with n vertices and c connected components. Write its components as

$$T_1, T_2, \dots, T_c.$$

Each component T_i is a tree, so it has

$$|V(T_i)| - 1$$

edges. Summing over all components gives

$$m = \sum_{i=1}^c (|V(T_i)| - 1) = \sum_{i=1}^c |V(T_i)| - \sum_{i=1}^c 1 = \sum_{i=1}^c |V(T_i)| - c = n - c.$$

Now assume that G is acyclic and $m = n - 1$. Since G is acyclic, it is a forest. By the formula above, $m = n - c$. But we are also assuming that $m = n - 1$, so

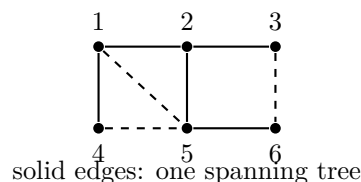
$$n - c = n - 1.$$

We obtain $c = 1$. In other words, G has exactly one connected component, which means G is connected. This proves the implication

$$(3) \implies (1).$$

Combining the three implications, we have proved that the three conditions above are equivalent.

Spanning trees. Let G be a connected graph. A **spanning tree** of G is a subgraph $T \subseteq G$ such that $V(T) = V(G)$ and T is a tree. Basically, a spanning tree keeps all the vertices but only enough edges to remain connected.



Every connected graph has a spanning tree. To see this, start with a connected graph and delete edges from cycles one at a time until no cycles remain. Deleting an edge from a cycle does not disconnect the graph, and we never delete vertices. The final graph is connected and acyclic, so it is a spanning tree.

Deleting one vertex. Here is a useful application of spanning trees.

Theorem. If G is connected and $|V(G)| \geq 2$, then there is a vertex v such that $G - v$ is connected.

Proof. Choose a spanning tree T of G . Since T has at least two vertices, it has at least two leaves. Choose a leaf v of T . By the leaf-removal fact for trees, $T - v$ is connected. Since

$$T - v \subseteq G - v,$$

any two vertices in $G - v$ are already connected by paths inside $T - v$. In particular, the graph $G - v$ is connected. $\square$

The previous result demonstrates that we can always carefully prune one vertex from a connected graph without breaking connectedness.

Dominating sets. A set $D \subseteq V(G)$ is a **dominating set** if for every vertex $v \in V(G)$, either $v \in D$ or v has a neighbor in D .

Theorem. If G is connected and $n \geq 2$, then G has a dominating set of size at most $\frac{n}{2}$.

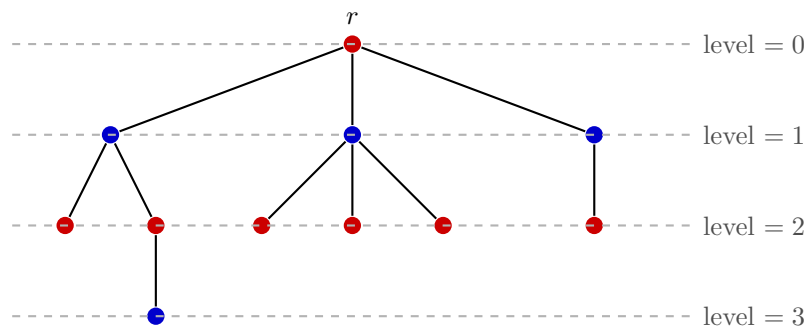
Proof. Choose a spanning tree T of G and root it at a vertex r . The root is on level 0. Its children are on level 1, their children are on level 2, and so on.

Define two sets according to level:

$$A = \{v \in V(T) : v \text{ is on an even level}\},$$

and

$$B = \{v \in V(T) : v \text{ is on an odd level}\}.$$



The set A is a dominating set in T . Indeed, every vertex in A is already in the set, and every vertex in B has a parent in A .

Similarly, B is also a dominating set in T . Every vertex in B is already in the set. Also, every vertex in A has a neighbor in B : the root has a child because $n \geq 2$, and every other vertex in A has a parent in B .

Since

$$|A| + |B| = n,$$

the smaller of the two sets has size at most $\lfloor n/2 \rfloor$. Since both A and B are dominating sets in T , the smaller one is a dominating set in T . Since $T \subseteq G$, the same set is also a dominating set in G . $\square$

Bipartite graphs and odd cycles. A graph is **bipartite** if its vertex set can be written as

$$V(G) = A \cup B,$$

where A and B are disjoint and every edge has one endpoint in A and the other endpoint in B .

Theorem. A graph G is bipartite if and only if G has no odd cycles.

Proof. First, suppose G is bipartite with bipartition $V(G) = A \cup B$. Any cycle must alternate sides:

$$A, B, A, B, \dots$$

To return to the starting vertex, the cycle must have even length. This shows that a bipartite graph has no odd cycles.

Conversely, suppose G has no odd cycles. It is enough to argue component by component. In each connected component, choose a spanning tree T and root it at some vertex r . Put the vertices at even distance from r into A , and put the vertices at odd distance from r into B .

We claim that no edge of G joins two vertices on the same side. Suppose, to the contrary, that there is an edge uv whose endpoints are on the same side. Then u and v have the same parity of depth in the rooted spanning tree T . The unique path from u to v in T therefore has even length. Adding the edge uv to that path creates a cycle of odd length, a contradiction.

Since every edge goes between A and B , we conclude that G is bipartite. $\square$

As an immediate consequence, every tree is bipartite, since a tree has no cycles at all.